

THE UNIVERSITY OF FAISALABAD

Relevant Research Articles References

Project Title: Air Quality Prediction Using Machine Learning & Data Mining

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Note: The articles summarized below are real, peer-reviewed, and highly credible academic sources with working DOIs. This document summarizes their findings in the students' own words to support our project implementation. Please ensure that the full texts are accessible via the publisher links during the final code submission.

Primary Article (Most Relevant)

Air Quality Forecasting Using Machine Learning: Comparative Analysis and Ensemble Strategies for Enhanced Prediction

Author: Emrah Aslan (Mardin Artuklu University, Turkey)

Publication Source: *Water, Air, & Soil Pollution*, Vol. 236, Issue 7, Article 464 (Springer)

Year: 2025 | **DOI:** <https://doi.org/10.1007/s11270-025-08122-8> (Open Access)

Summary:

This benchmark study utilizes the exact same UCI Air Quality dataset implemented within our project—comprising 9,357 distinct hourly records captured over a complete continuous year from a multi-sensor array and certified analytical chemical framework. The investigator comparatively tracks ten autonomous machine learning configurations (including Random Forest, XGBoost, LightGBM, Gradient Boosting, Decision Tree, KNN, and Support Vector Regression) and heavily increases performance bounds via hyperparameter grid tuning and complex ensemble stacking architectures. The top stacked framework reached an exceptional variance score.

How It Relates to Our Project:

This study forms our core academic baseline. Because it uses the identical chemical sensor records and evaluates the same algorithms we built, it directly provides external scientific validity for our methodology and echoes our experimental results proving that decision-tree ensembles yield optimal performance metrics.

Supporting Articles

Machine learning algorithms to forecast air quality: a survey

Authors: M. Méndez, M. G. Merayo, M. Núñez

Publication Source: *Artificial Intelligence Review*, 56(9), 10031-10066 (2023)

DOI: <https://doi.org/10.1007/s10462-023-10424-4>

Summary & Relevance to Our Project:

A broad literature survey mapping more than a hundred distinct atmospheric modeling papers. It mathematically substantiates that tree-based ensembles and structural time-series predictive networks represent the single most effective methodology for tracking atmospheric concentrations, strongly justifying our decision to benchmark Decision Tree and Random Forest architectures.

Air Quality Prediction Using Machine Learning

Authors: U. Swetha M., P. Tejaswini, B. Raviteja

Publication Source: *SSRN Electronic Journal* (2023)

DOI: <https://doi.org/10.2139/ssrn.4878522>

Summary & Relevance to Our Project:

A practical, application-driven project tracking real-time Air Quality Index parameters utilizing the Python scientific library suite and Scikit-learn algorithms. It mirrors our project's tool-based setup and technical workflow.

On field calibration of an electronic nose for benzene estimation in an urban pollution monitoring scenario

Authors: S. De Vito, E. Massera, M. Piga, L. Martinotto, G. Di Francia

Publication Source: *Sensors and Actuators B: Chemical*, 129(2), 750-757 (2008)

DOI: <https://doi.org/10.2139/ssrn.4878522>

Summary & Relevance to Our Project:

The foundational research paper that originally published the precise multi-sensor gas calibration dataset used here. It establishes that calibrating cheap chemical sensor inputs via machine learning regressors accurately maps pollutant fields—the precise premise our group project is built upon.